

AGGREGATE TRENDS

PRESENTATION 1

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Monday, October 13th



Our Agenda

01	New Metric Ideas
02	Dashboard Development
03	Visual Insights & Business Implications
04	Next Steps
05	Appendix

New Metric Ideas

Potential Metric	What it Measures	Why it Matters
Submenu Efficiency	Average number of menu steps before a failed outcome (client has to call back)	Shows how deep callers travel before failure to identify path inefficiencies
Redirection Performance	Measures how often redirected calls reach a live staff member (human)	Shows how effectively callers are connected to real humans during the redirection process to identify potential staffing or routing gaps
First-Route Success Rate	Measures how many calls are answered on the first attempt without redirection	Captures how often callers reach the right staff member immediately to reflect how efficient routes currently are
Self-Service Success Rate	Measure of how many calls that never reach a live agent do not end with the customer abandoning the call	Shows the effectiveness of the automation system in helping customers to identify ways to reduce the call load on live agents

Features to be Displayed on the Dashboard

Graphs



1

Call Duration Histogram

2

Menu Usage Treemap

3

Path Depth Histogram

Statistics



4

Outcome Breakdown

5

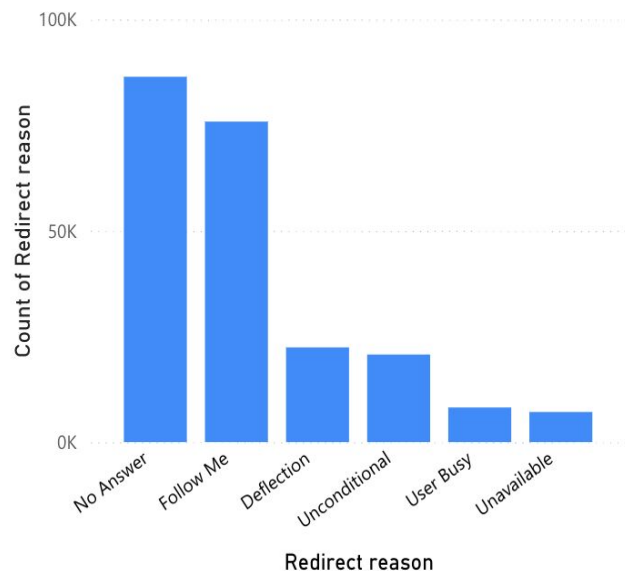
Submenu Efficiency Metric

6

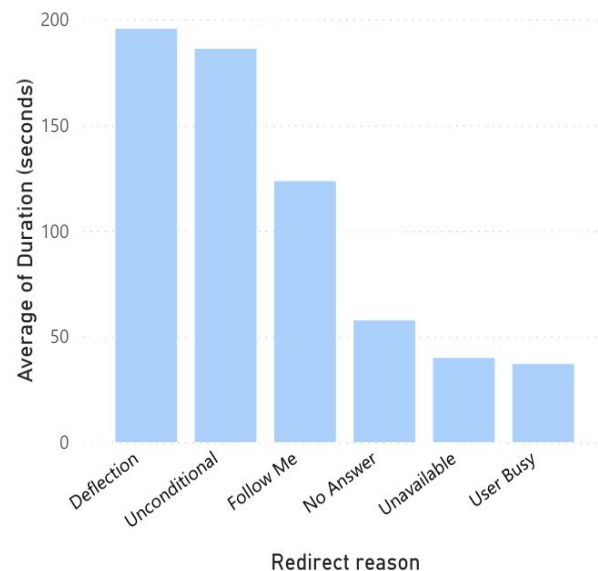
Redirection Performance Metric

Redirection Performance (April 2024 - August 2025)

Volume of Redirection Events by Redirect Type



Average of Duration per Redirection Type (seconds)



What We Found:

- **Follow Me and No Answer** are the most common redirect reasons; many callers don't connect on the first try
- **Automated transfers (like Deflection and Unconditional) keep callers on the line longer**, possibly reaching voicemail or holding queues
- **Shorter "Unavailable" and "User Busy" calls** may indicate staff capacity limits or missed call handling

Note: This analysis focuses on inbound (terminating) calls, since these represent clients reaching out for help, this filter was applied on the dashboard. Because a single call can include multiple legs under one Correlation ID, these visuals reflect the volume and duration of redirection activity rather than the total number or duration of complete calls.

Improving Client Connection

1

Review call routing setup

Audit Follow Me and forwarding rules to reduce unnecessary personal redirects and improve first-contact response rates.



2

Strengthen intake availability

Use call volume data to adjust staffing during “No Answer” periods to maximize availability for answering core lines.



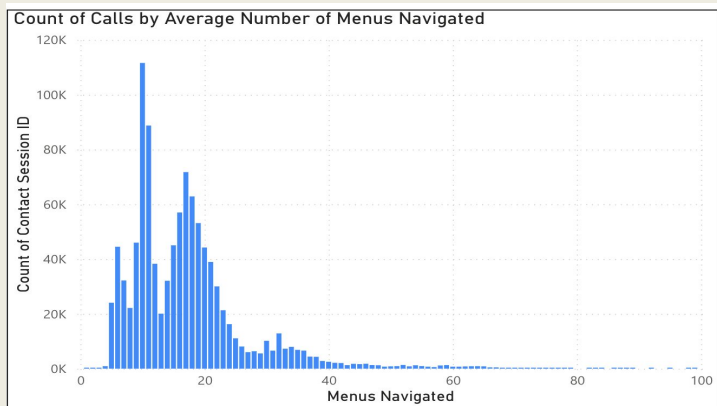
3

Simplify redirection paths

Prioritize routing callers directly to live staff or streamlined automated queues.

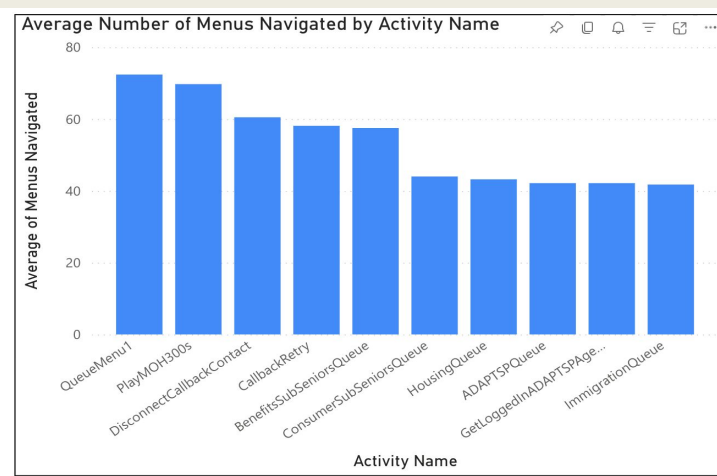


Submenu Efficiency



17.74

Average of Menus Navigated



What We Found:

- Many clients navigate through **long and complex menus** before reaching help
- Some calls paths (like QueueMenu1 and PayNH030186) **show extremely high menu counts**, suggesting they may be confusing callers
- **Disconnection and callback activities** also appear frequently among call paths which could imply abandonment before connecting to staff

Simplifying the Caller Journey

1

Audit long menu paths

Review and re-map call flows for menus showing above-average navigation counts to identify the redundant and confusing steps



2

Simplify top-level menus

Reduce the number of nested options and ensure high-demand services (like intake or benefits) are available within one or two steps



3

Evaluate activities with complex paths

Observe which activities are causing the most submenu usage, and explore options as to how to simplify these activities



Next Steps & Further Questions

- **Validate call outcomes:** Determine whether longer call durations represent productive conversations or client confusion.
- **Deepen redirection analysis:** Explore why “Unconditional” and “Deflection” calls last longer and whether they reflect effective routing or delays.
- **Track multiple redirections:** Examine if certain redirect reasons lead to repeated transfers or dropped calls.
- **Validate menu depth:** Determine whether higher submenu counts reflect necessary complexity for certain activities or unnecessary caller confusion.
- **Simplify navigation paths:** Explore opportunities to streamline or merge low-value submenus to reduce average menu depth and improve caller experience.

Appendix

** Includes additional details and visuals from our preliminary analysis that were not included in the live presentation due to time constraints.*



Data Cleaning (Power BI)

The following was completed in **Power BI**.

- **All Calls Dataset**

- Combined multiple All-Calls files into one dataset using import script from Professor Krish
- Called Number → Converted to Text to handle special characters like “#”
- Applied Transform → Format → Trim to the Redirect Reason column to remove extra spaces
- Renamed inconsistent labels for readability (e.g., “NoAnswer” → “No Answer”)
- Removed all rows where Duration ≤ 0 , as these represented failed or incomplete calls

- **CAR Dataset**

- Created navigation variable using COUNTIF formula to calculate number of menus navigated per call

Data Cleaning (Python)

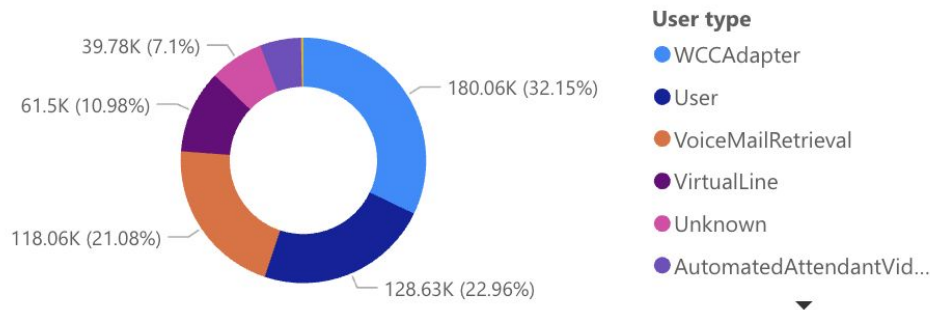
The following data cleaning was completed in **Python**. Code [here](#).

- **All Calls Dataset**

- Turned all columns with mixed data types to strings
- Removed all columns that contained exclusively NaNs
- Converted all columns with phone numbers into strings rather than int/float
- Grouped all data by Correlation ID
- Filtered the data to only inbound customer calls

Who Answers Legal Aid's Calls?

Recipients of All Calls (Terminating)



Our Recommendation:

Streamline voicemail handling and reduce unnecessary voicemail routing

A high share of calls end in voicemail which creates a bottleneck for both clients and staff. Optimizing routing and voicemail processes could significantly reduce response time and improve access to the legal help callers need.



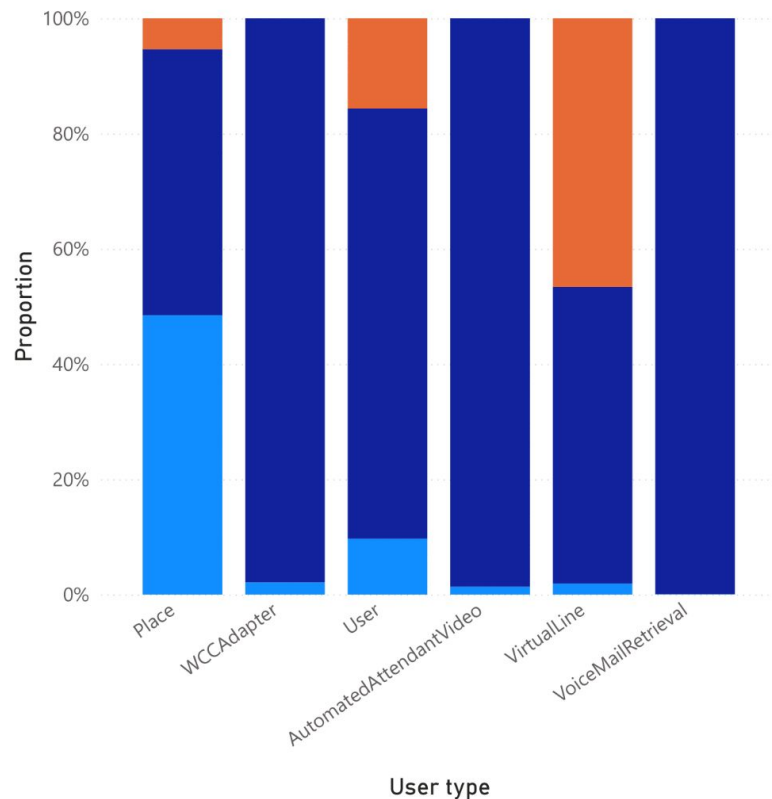
What We Found:

- **Most inbound calls go to automated systems** showing that a large share of clients interact with technology rather than a live staff member.
- **Only about one in four calls (23%) reach a direct “User”**, meaning many callers may not connect to a person on their first try.
- **A smaller share (≈18%) end at Virtual Lines or Unknown users**, suggesting possible data gaps where follow-up accountability may be unclear

Proportion of Answer Indicators by User Type

Proportion of Answer Indicators By User Type

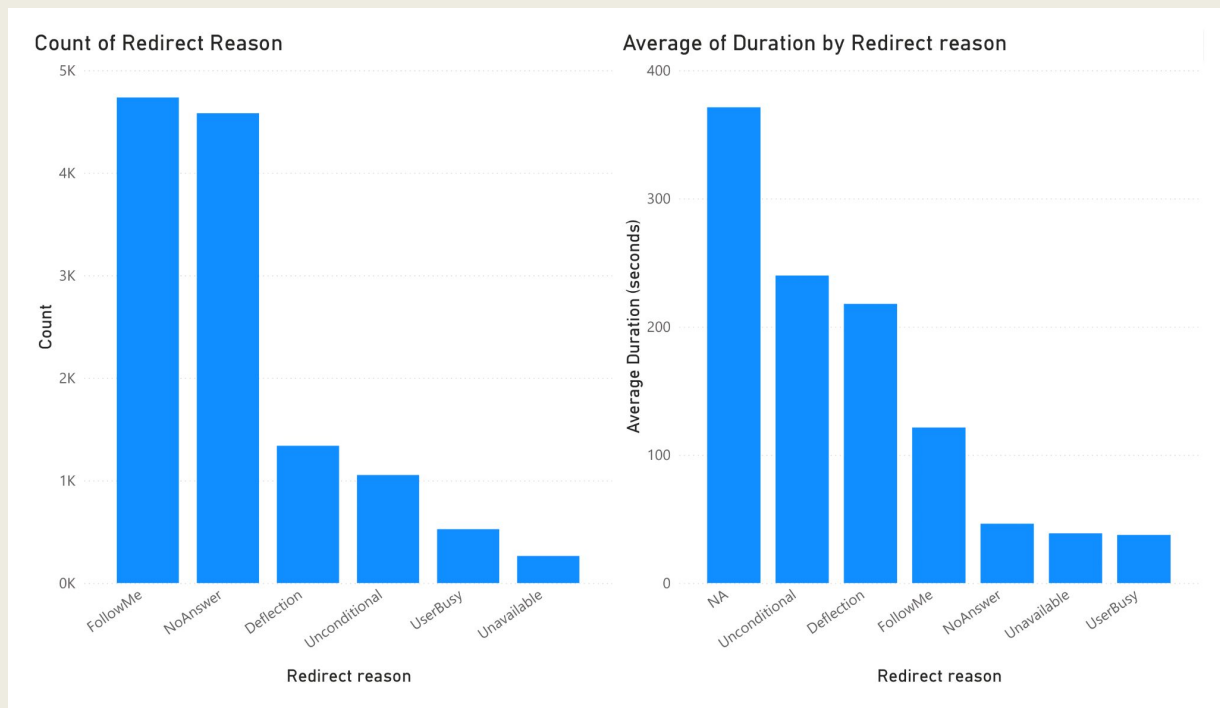
Answer Indicator ● No ● Yes ● Yes-PostRedirection



Business Impact

- **Most user types (especially WXCAdapter, VirtualLine, and VoiceMailRetrieval) show nearly all calls as answered directly**
- **Place and AutomatedAttendantVideo show notable “No” rates, meaning a large share of calls go unanswered.**
- **User and VirtualLine show some “Yes-PostRedirection”, meaning calls are often rerouted before reaching the right person**

Redirection Performance (August 2025)



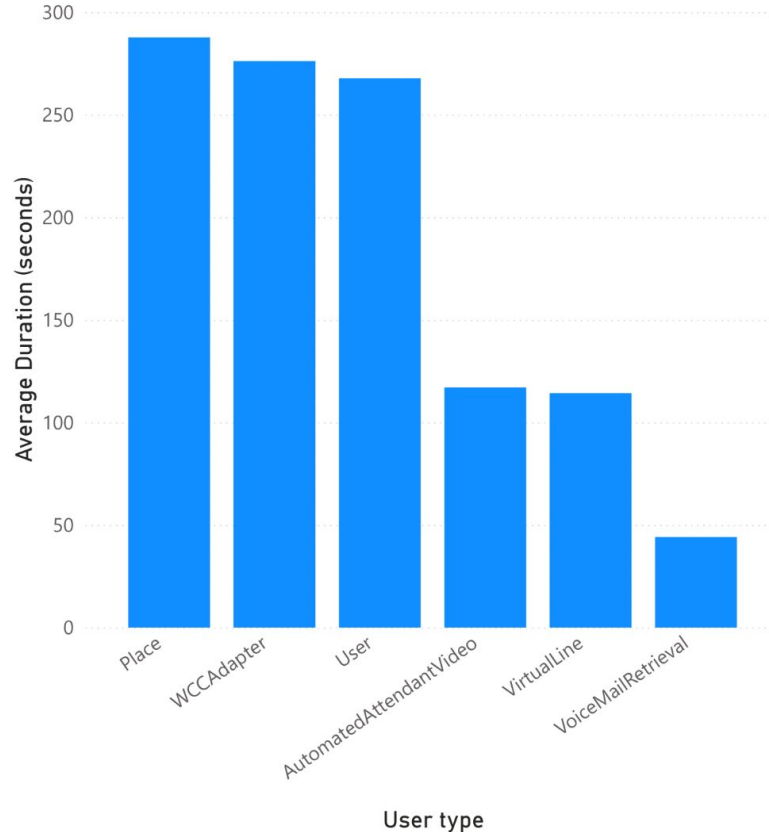
Business Impact

- “Deflection” & “Unconditional” calls are less common but show **longer durations**
- “Unconditional” → **point of improvement** with Call Forward Always CFA service?
- “Deflection” → **point of investigation** for causes
- Calls labeled “FollowMe” and “NoAnswer” occur most frequently but have **shorter call durations** → **point of investigation for FollowMe**

Note: This analysis focuses on inbound (terminating) calls, since these represent clients reaching out for help, this filter was applied on the dashboard. Only August 2025 data was used to provide a recent snapshot of redirection activity and call handling performance. This preliminary view helps identify whether current redirection patterns differ from longer-term trends across the past year.

Average Duration by User Type

Average Duration by User Type

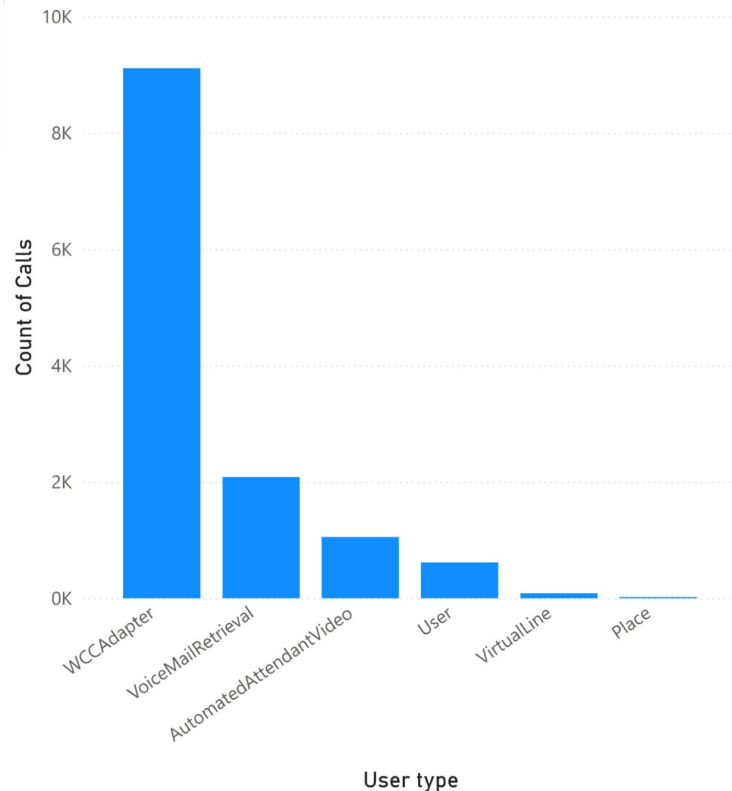


Business Impact

- Calls handled by “Place,” “WXCAdapter,” and “User” last the longest
- Automated and virtual systems (AutomatedAttendantVideo, VirtualLine) show shorter durations, possibly indicating quick routing or early disconnections
- Voicemail calls are very short, showing that when calls go to voicemail, clients potentially hang up quickly

Count of End Leg User Types For Customer Calls

Count of End Leg User Type For Customer Calls



Business Impact

- **User** (calls ending with a live agent) is very low; the vast majority of customer calls do not end with a live agent
- **VoiceMailRetrieval** is the 2nd highest; there are more calls that result in voicemail than calls that result in live agents
- Most calls ending in **WCCAdapter** suggests that most abandonments occur in-queue